



A02 PAT and AOI communication spec.

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Engineer / MAMBA

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修改日誌

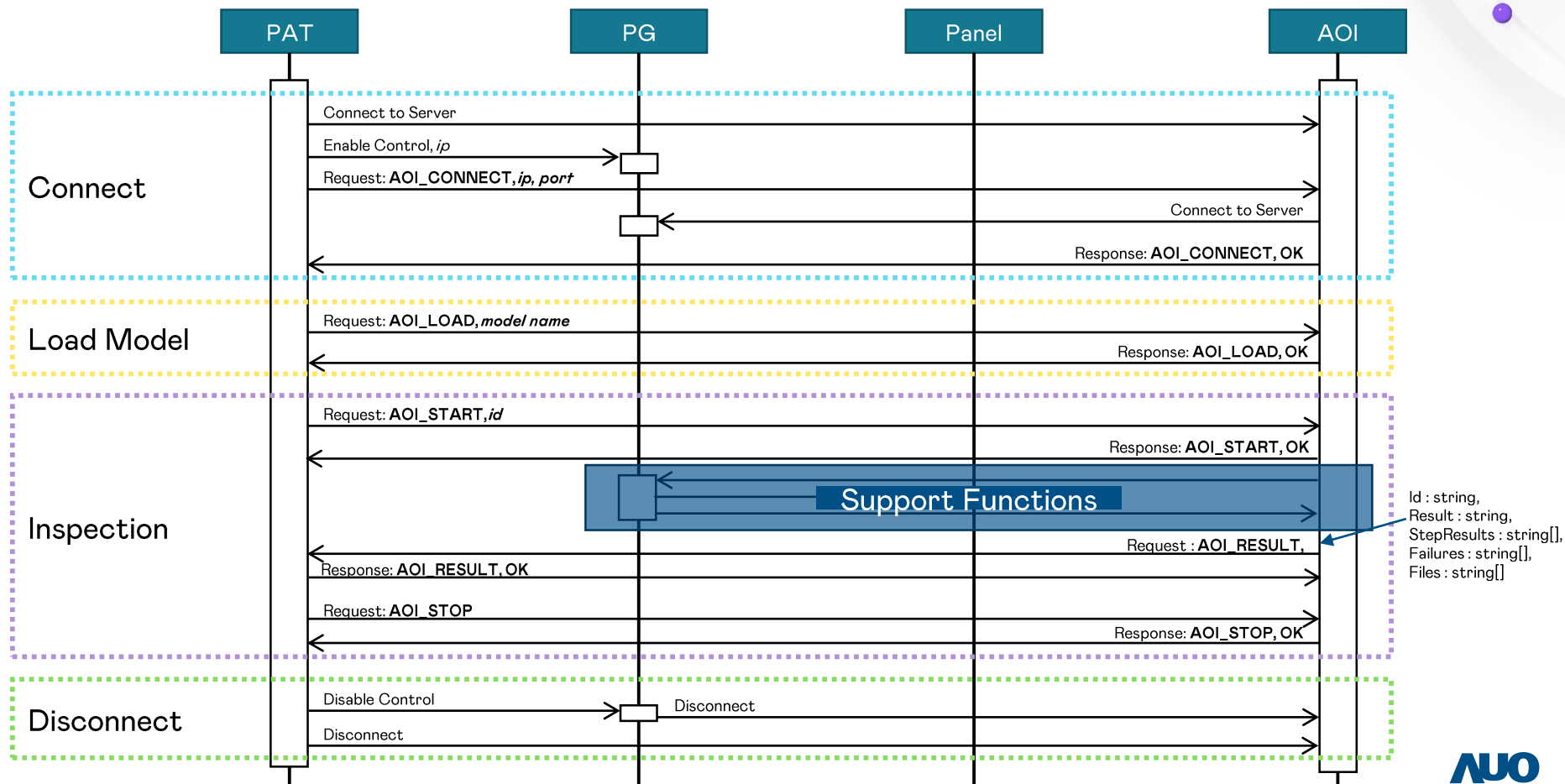
版本	日期	作者	修改內容
V2025.12.01	2025.12.01	YHChen	1. 初版
V2026.01.17	2026.01.17	YHChen	1. 修正通訊套件 (AssC → EolLinker) 2. 新增支援 Symbol Insepction 功能
V2026.08.25	2026.08.25	JohnnyTsai	1. 新增 AOI_RESULT / AOI_SETPOSITION 2. AOI_SETPATTERN改為SETMODE 3. 修改 AOI_START格式

Hardware



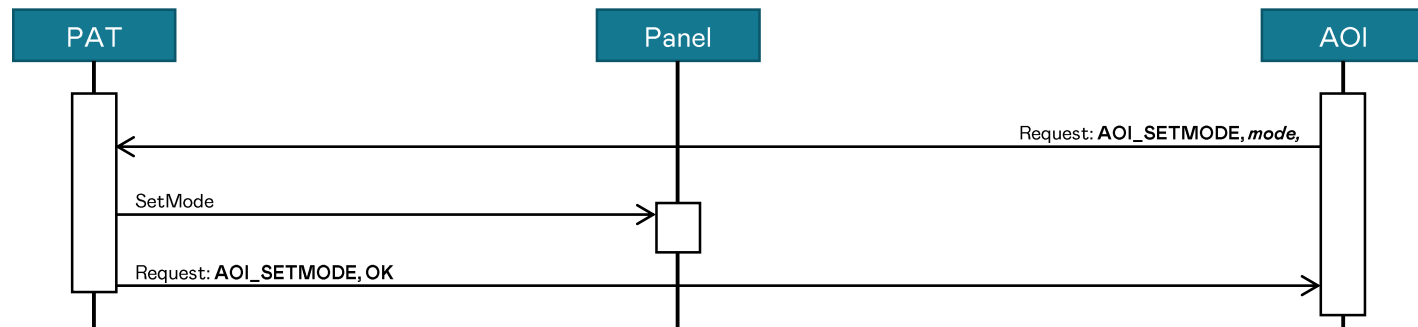
通訊流程

AOI 流程

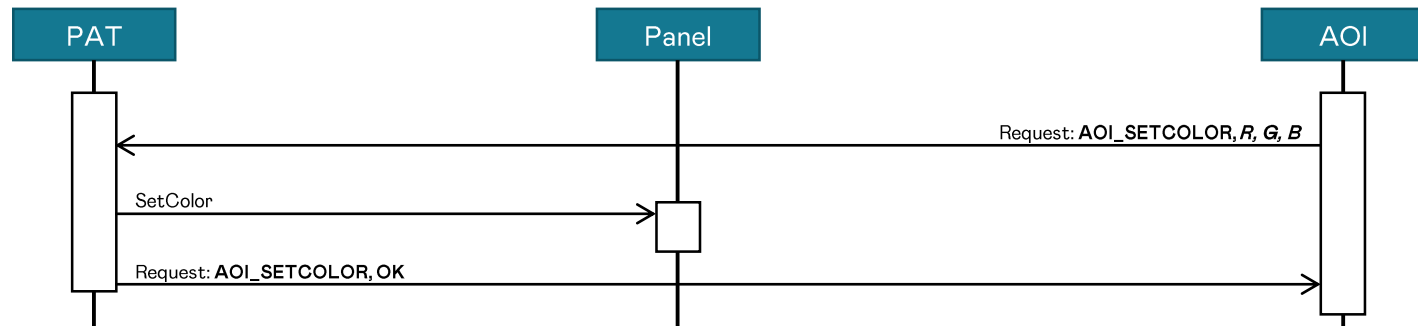


輔助功能

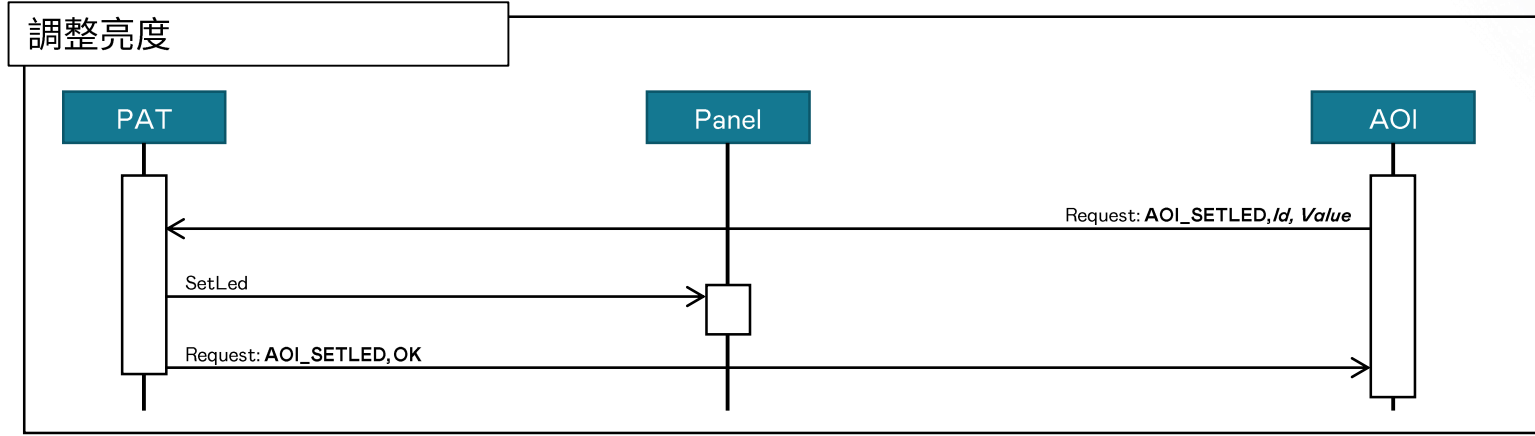
切換模式



切換純色畫面



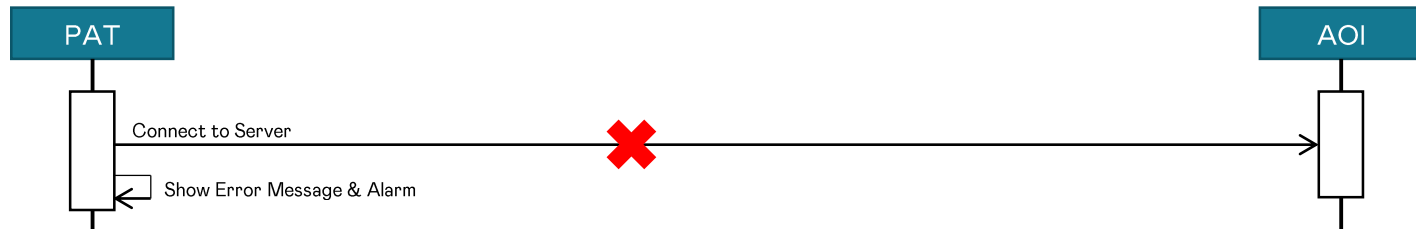
輔助功能



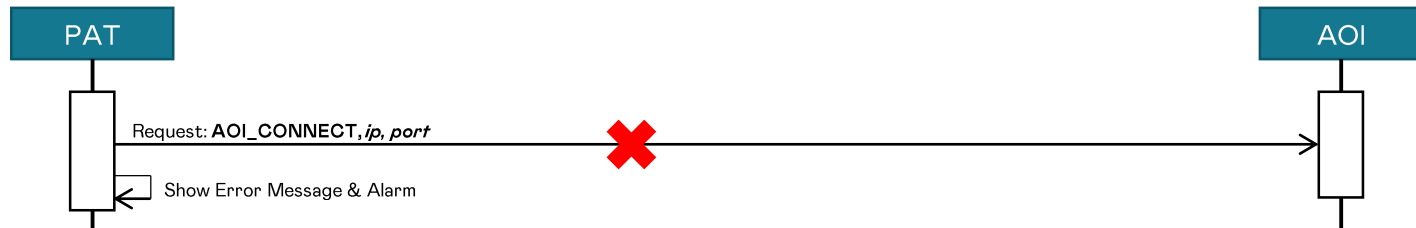
異常狀況



PAT 無法建立連線

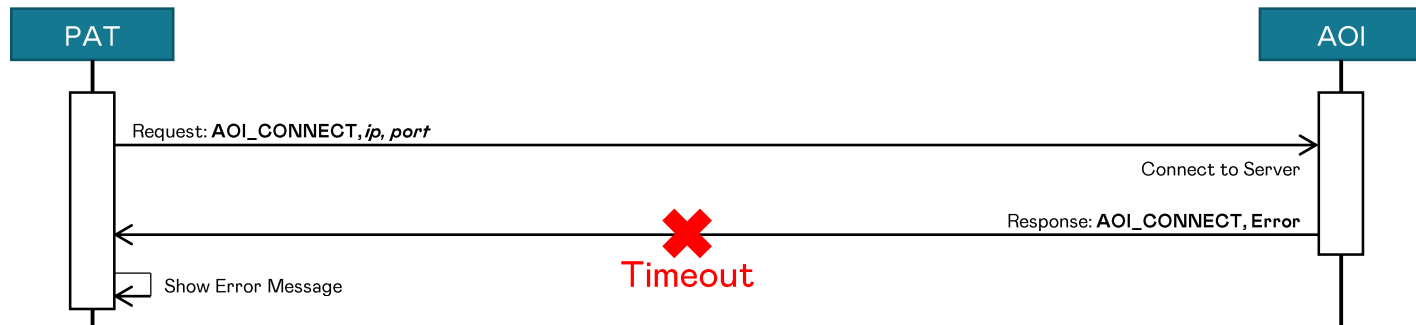


PAT 無法發送命令

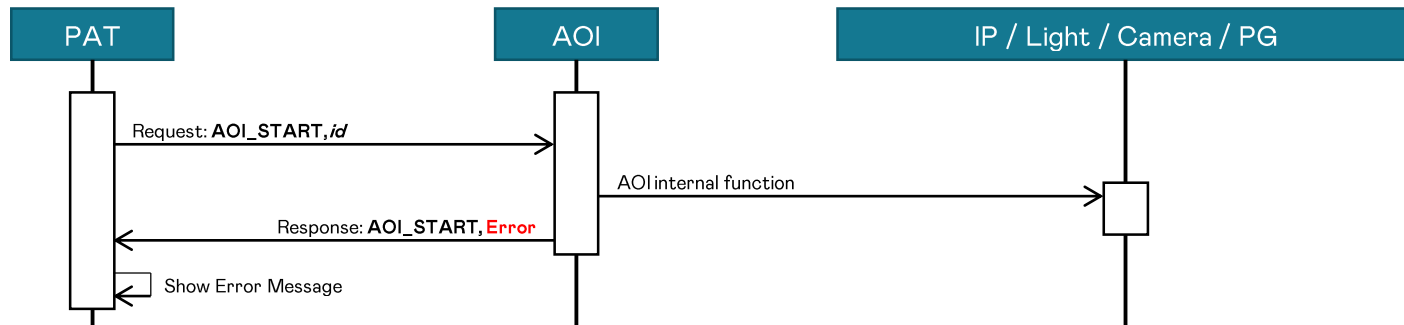


異常狀況

PAT 請求超時



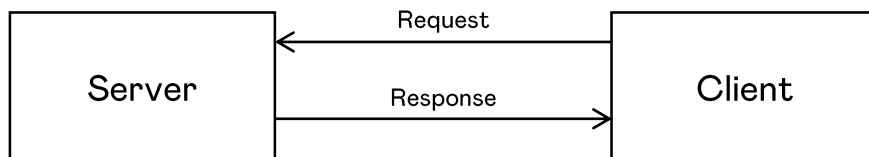
PAT 收到錯誤



通訊命令

通訊命令

- 使用 EolLinker.dll，走 TCP/IP Server/Client 架構，傳送 json 格式



Request Data Json

```
{
  "Command": "some_command_name",
  "Parameters": {
    "Key1": [
      "ValueA",
      "ValueB"
    ],
    "Key2": [
      "ValueC"
    ]
  }
}
```

Response Data Json

```
{
  "Command": "some_command_name",
  "Result": "Ok",
  "Parameters": {
    "Key1": [
      "ValueA",
      "ValueB"
    ],
    "Key2": [
      "ValueC"
    ]
  }
}
```

Result List:

None	→ 沒設定
Ok	→ 沒問題
Ng	→ 未通過
Error	→ 命令執行時出現錯誤
Timeout	→ 指令超時

通訊命令 (client : PAT → server : AOI)

- PAT 呼叫 AOI 進行檢測動作

需求描述	命令 與 結果	參數	備註
連線確認	AOI_CONNECT	Ip : string, Port : string	
切換設定檔	AOI_LOAD	Model : string	
開始檢測	AOI_START	Id : string	
停止動作	AOI_STOP		

通訊命令 - 連線確認 (AOI_CONNECT)

<Description> : Client ask Server to connect to PatternGen.

<Request Command> : AOI_CONNECT

<Request Parameters>

<Key & Value> Ip : string, TCP/IP format

<Key & Value> Port: string, 1 ~ 65535

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{
  "Command": "AOI_CONNECT",
  "Parameters": {
    "Ip": [ "127.0.0.1" ],
    "Port": [ "9527" ]
  }
}
```

Response OK

```
{
  "Command": "AOI_CONNECT",
  "Result": "Ok",
  "Parameters": {}
}
```

Response Error

```
{
  "Command": "AOI_CONNECT",
  "Result": "Error",
  "Parameters": {
    "ErrorMessage": [ "Connect Failed" ],
    "ErrorCode": [ "3125" ]
  }
}
```

Response Timeout

```
{
  "Command": "AOI_CONNECT",
  "Result": "Timeout",
  "Parameters": {}
}
```

通訊命令 - 切換設定檔 (AOI_LOAD)

<Description> : Switch to new Model recipe.

<Request Command> : **AOI_LOAD**

<Request Parameters>

<Key & Value> Model : string

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{
  "Command": "AOI_LOAD",
  "Parameters": {
    "Model": [ "Default" ] ]
}
```

Response OK

```
{
  "Command": "AOI_LOAD",
  "Result": "Ok",
  "Parameters": {}
}
```

Response Error

```
{
  "Command": "AOI_LOAD",
  "Result": "Error",
  "Parameters": {
    "ErrorMessage": [ "No Model" ]
    "ErrorCode": [ "5927" ]
  }
}
```

Response Timeout

```
{
  "Command": "AOI_LOAD",
  "Result": "Timeout",
  "Parameters": {}
}
```

通訊命令 - 開始檢測 (AOI_START)

<Description> : Start to running inspect.

<Request Command> : AOI_START

<Request Parameters>

<Key & Value> Id : string

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{
  "Command": "AOI_START",
  "Parameters": {
    "Id": ["0123456"]
  }
}
```

Response OK

```
{
  "Command": "AOI_START",
  "Result": "Ok",
  "Parameters": {}
}
```

Response Error

```
{
  "Command": "AOI_START",
  "Result": "Error",
  "Parameters": {
    "ErrorMessage": ["ID Not Found"],
    "ErrorCode": ["5927"]
  }
}
```

Response Timeout

```
{
  "Command": "AOI_START",
  "Result": "Timeout",
  "Parameters": {}
}
```

通訊命令 - 停止動作 (AOI_STOP)

<Description> : Stop everything and reset statue.

<Request Command> : AOI_STOP

<Request Parameters>

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{  
  "Command": "AOI_STOP",  
  "Parameters": {}  
}
```

Response OK

```
{  
  "Command": "AOI_STOP",  
  "Result": "Ok",  
  "Parameters": {}  
}
```

Response Error

```
{  
  "Command": "AOI_STOP",  
  "Result": "Error",  
  "Parameters": {  
    "ErrorMessage": ["Unexpected Error"]  
    "ErrorCode": ["9453"]  
  }  
}
```

Response Timeout

```
{  
  "Command": "AOI_STOP",  
  "Result": "Timeout",  
  "Parameters": {}  
}
```

通訊命令 (client : AOI → server :PAT)

- AOI 呼叫 PAT 進行檢測動作

需求描述	命令 與 結果	參數	備註
切換圖樣	AOI_SETMODE	Pattern : enum DAY/NIGHT	
移動機構	AOI_SETPOSITION	-	切換至 Knob 位置
上報結果	AOI_RESULT	Id : string, Result : string, StepResults : string[], Failures : string[], Files : string[]	
切換純色	AOI_SETCOLOR	Red : int 0 ~ 255, Green : int 0 ~ 255, Blue : int 0 ~ 255	保留功能
調整LED亮度	AOI_SETLED	Id : int 0 ~ 255, Value : double -100 ~ 100	保留功能

通訊命令 - 切換圖樣 (AOI_SETMODE)

<Description> : Switch to new Pattern.

<Request Command> : AOI_SETMODE

<Request Parameters>

<Key & Value> Mode : enum DAY / NIGHT

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{
  "Command": "AOI_SETMODE",
  "Parameters": {
    "Mode": ["NIGHT"],
  }
}
```

Response OK

```
{
  "Command": "AOI_SETMODE",
  "Result": "Ok",
  "Parameters": {}
}
```

Response Error

```
{
  "Command": "AOI_SETMODE",
  "Result": "Error",
  "Parameters": {
    "ErrorMessage": ["Mode not Found"]
  },
  "ErrorCode": ["9487"]
}
```

Response Timeout

```
{
  "Command": "AOI_SETMODE",
  "Result": "Timeout",
  "Parameters": {}
}
```

通訊命令 - 切換位置(AOI_SETPOSITION)

<Description> : Switch to Knob Position.

<Request Command> : AOI_SETPOSITION

<Request Parameters>

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

<Example>

Request Data

```
{  
  "Command": "AOI_SETPOSITION",  
  "Parameters": {}  
}
```

Response OK

```
{  
  "Command": "AOI_SETPOSITION",  
  "Result": "Ok",  
  "Parameters": {}  
}
```

Response Error

```
{  
  "Command": "AOI_SETPOSITION",  
  "Result": "Error",  
  "Parameters": {  
    "ErrorMessage": ["Move Error"]  
    "ErrorCode": ["9453"]  
  }  
}
```

Response Timeout

```
{  
  "Command": "AOI_SETPOSITION",  
  "Result": "Timeout",  
  "Parameters": {}  
}
```

通訊命令 - 上報結果 (AOI_RESULT)

<Description> : Send All Inspect Result. (Include Day Mode / Night Mode / Knob)

<Request Command> : AOI_RESULT

<Request Parameters>

<Key & Value> Id : string

<Key & Value> Result : string

<Key & Value> StepResults : string[]

<Key & Value> Failures : string[]

<Key & Value> Files : string[]

<Response Result>

Ok : Request well done.

Error : Something happened in Server. Response should return error code and error message.

Timeout : Network Communication Timeout.

通訊命令 - 上報結果 (AOI_RESULT)

<Example>

Request Data

```
{
  "Command": "AOI_RESULT",
  "Parameters": {
    "Id": ["01234567"],
    "Result": ["OK"],
    "StepResults": ["OK", "OK", "OK"],
    "Failures": [],
    "Files": ["Day_01234567.jpg", "Night_01234567.jpg", "Knob_01234567.jpg"]
  }
}
```

Request Data

```
{
  "Command": "AOI_RESULT",
  "Parameters": {
    "Id": ["01234567"],
    "Result": ["NG"],
    "StepResults": ["OK", "NG", "OK"],
    "Failures": ["Night Mode: Color out of specification"],
    "Files": ["Day_01234567.jpg", "Night_01234567.jpg", "Knob_01234567.jpg"]
  }
}
```

Response OK

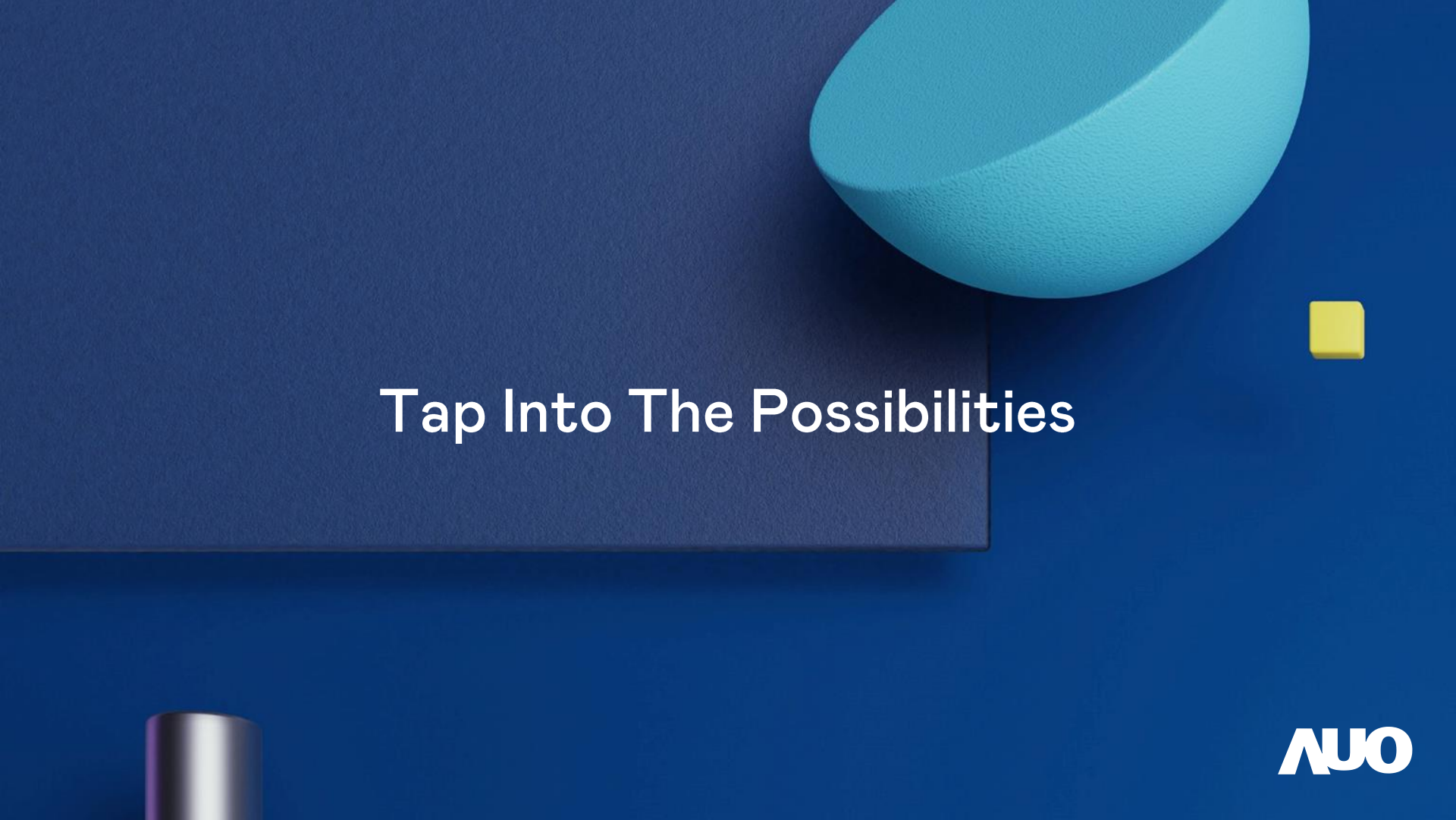
```
{
  "Command": "AOI_RESULT",
  "Result": "Ok",
  "Parameters": {}
}
```

Response Error

```
{
  "Command": "AOI_RESULT",
  "Result": "Error",
  "Parameters": {
    "ErrorMessage": ["Failed to save result to database."],
    "ErrorCode": ["5001"]
  }
}
```

Response Timeout

```
{
  "Command": "AOI_RESULT",
  "Result": "Timeout",
  "Parameters": {}
}
```



Tap Into The Possibilities

AUO